



IT - ITeS SSC
NASSCOM



Qualification Pack



Application Developer - Web & Mobile

Electives: Front-end Web Development/ Mobile Application Development/
Back-end Engineering

QP Code: SSC/Q8403

Version: 3.0

NSQF Level: 5

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SSC/Q8403: Application Developer - Web & Mobile

Brief Job Description

An Application Developer - Web and Mobile designs, develops, and maintains innovative applications that fulfill user needs and business objectives. The role involves managing the entire application lifecycle, ensuring efficient workflows for continuous integration and delivery. The individual collaborates with cross-functional teams to gather requirements, conducts testing, and implements enhancements to optimize performance and user experience. The person also provides ongoing support for existing applications.

Personal Attributes

The individual should possess strong communication skills to collaborate effectively with different stakeholders. The person should have problem-solving skills, attention to detail, adaptability to changing requirements, and the ability to work independently and as part of a team. A proactive attitude and a passion for technology are also essential in the role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8417: Implement DevSecOps practices for secure application development](#)
2. [SSC/N8451: Implement Continuous Integration and Continuous Deployment \(CI/CD\) practices](#)
3. [SSC/N8125: Develop tests or simulations for end-to-end QA of systems](#)
4. [SSC/N8418: Fix application bugs and improve application performance](#)
5. [SSC/N8323: Monitor and manage cloud applications and the deployed systems](#)
6. [SSC/N8452: Optimize Revenue, Security, and Accessibility in Web and Mobile Products](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Front-end Web Development

This unit is about developing web application for different platforms keeping in mind the different requirements of the application/ solution

1. [SSC/N8414: Develop a user-friendly web app aligned with functional, non-functional, and UX requirements](#)

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Elective 2: Mobile Application Development

This unit is about developing mobile applications for different types of mobile platforms

1. [SSC/N8415: Develop native/cross-platform/hybrid mobile application for the target platforms](#)

Elective 3: Back-end Engineering

This unit is about developing reliable and secure back-end APIs and services for any software product/ solution

1. [SSC/N8416: Develop reliable, scalable and secure back-end aligned to the application architecture](#)

Qualification Pack (QP) Parameters

Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (UG or Diploma with courses related to Engg. / Science) OR Completed 3 year diploma after 10th (UG or Diploma with courses related to Engg. / Science) with 1.5 years of experience In relevant field. Relevant Experience (including work, internship and apprenticeship) related to Web and Mobile OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience in relevant field Relevant Experience (including work, internship and apprenticeship) related to Web and Mobile
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA



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Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	3.0
Reference code on NQR	QG-05-IT-03642-2025-V2-NASSCOM
NQR Version	3.0

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SSC/N8417: Implement DevSecOps practices for secure application development

Description

This unit is about integrating security into the application development lifecycle through DevSecOps practices. It includes managing secure code repositories, ensuring security compliance during development, and conducting security assessments to identify vulnerabilities

Scope

The scope covers the following :

- Secure version control and source code management
- Security compliance and auditing
- Security in deployment pipelines

Elements and Performance Criteria

• *Secure version control and source code management*

To be competent, the user/individual on the job must be able to:

- PC1.** maintain and secure the repository, ensuring appropriate access controls and backup procedures for managing application source code
- PC2.** manage changes to the application code/source code through a version control system, tracking all changes and contributions
- PC3.** implement policies for code review, tagging, and secure merging to minimize vulnerabilities in the development process
- PC4.** integrate version control systems with project management tools, streamlining workflow and collaboration across teams

Security compliance and auditing

To be competent, the user/individual on the job must be able to:

- PC5.** conduct security reviews of application source code, including vulnerability scanning and static code analysis
- PC6.** integrate security tools into the DevSecOps pipeline to automate security checks throughout the development lifecycle
- PC7.** document security practices and maintain compliance with industry standards (e.g., ISO/IEC 27001)

Security in deployment pipelines

To be competent, the user/individual on the job must be able to:

- PC8.** ensure security measures are applied during deployment, such as encryption, access management, and firewall configuration
- PC9.** conduct penetration testing and security audits before and after deployment to identify and mitigate risks

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** How to continuously integrate changes to application source code
- KU2.** the procedures and guidelines related to developing roadmaps for products/services
- KU3.** the policies and procedures for documenting roadmaps and collecting business requirements
- KU4.** the competitor product analysis techniques to inform product differentiation strategy
- KU5.** how to analyze and benchmark products and offerings of competitors
- KU6.** how to define and evaluate business requirements
- KU7.** how to analyze the impact of new products and features
- KU8.** how to develop product roadmaps.
- KU9.** how to identify and establish product metrics/KPIs
- KU10.** the best practices for managing secure repositories and handling sensitive data
- KU11.** the security tools for code analysis, vulnerability scanning, and encryption in the development lifecycle
- KU12.** Generative AI (Gen AI) and its applications
- KU13.** the use cases of Gen AI for product managers (e.g., project planning automation, user story generation)
- KU14.** prompt engineering techniques in Gen AI for defining prompts to generate relevant results

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** Work independently and collaboratively
- GS3.** communicate effectively in writing
- GS4.** evaluate business impact and share relevant information
- GS5.** analyze data and assess its implications for the business
- GS6.** relay relevant information to others
- GS7.** maintain attention to detail
- GS8.** ensure work is complete and error-free
- GS9.** adhere to rule-based decision-making processes
- GS10.** plan and organize work to meet targets and deadlines
- GS11.** apply problem-solving approaches in various situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
• <i>Secure version control and source code management</i>	15	25	-	8
PC1. maintain and secure the repository, ensuring appropriate access controls and backup procedures for managing application source code	3	7	-	2
PC2. manage changes to the application code/source code through a version control system, tracking all changes and contributions	2	6	-	2
PC3. implement policies for code review, tagging, and secure merging to minimize vulnerabilities in the development process	5	6	-	2
PC4. integrate version control systems with project management tools, streamlining workflow and collaboration across teams	5	6	-	2
<i>Security compliance and auditing</i>	10	15	-	7
PC5. conduct security reviews of application source code, including vulnerability scanning and static code analysis	3	5	-	3
PC6. integrate security tools into the DevSecOps pipeline to automate security checks throughout the development lifecycle	3	5	-	2
PC7. document security practices and maintain compliance with industry standards (e.g., ISO/IEC 27001)	4	5	-	2
<i>Security in deployment pipelines</i>	5	10	-	5
PC8. ensure security measures are applied during deployment, such as encryption, access management, and firewall configuration	3	5	-	3
PC9. conduct penetration testing and security audits before and after deployment to identify and mitigate risks	2	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8417
NOS Name	Implement DevSecOps practices for secure application development
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	5
Credits	3
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N8451: Implement Continuous Integration and Continuous Deployment (CI/CD) practices

Description

This unit focuses on the automation of the build, testing, and deployment of applications using CI/CD pipelines

Scope

The scope covers the following :

- Build and test automation
- CI/CD pipeline management
- Deployment automation

Elements and Performance Criteria

Build and test automation

To be competent, the user/individual on the job must be able to:

- PC1.** manage application environment variables and configurations for target environments
- PC2.** automate build and testing using scripts, tools, and test frameworks to ensure consistency and efficiency
- PC3.** integrate automated tests into the CI pipeline to detect errors early in the development cycle

CI/CD pipeline management

To be competent, the user/individual on the job must be able to:

- PC4.** implement and manage CI/CD pipelines, ensuring a smooth and automated process for continuous integration and delivery
- PC5.** test, identify, and fix build failures to ensure uninterrupted development and deployment cycles
- PC6.** manage deployment policies and processes, ensuring security and compliance are adhered to in the deployment pipeline

Deployment automation

To be competent, the user/individual on the job must be able to:

- PC7.** automate application deployment to different environments (e.g., web servers, API services, databases) using industry-standard tools (e.g., Jenkins, GitLab CI)
- PC8.** ensure deployment processes are scalable and maintain high availability, using containerization and orchestration tools (e.g. Docker, Kubernetes)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different product development life cycles (e.g., Waterfall, Agile, Lean) and their applicability
- KU2.** different data sources and how to access documents and information from data sources

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- KU3.** the techniques for identifying and evaluating product features based on market demand and customer feedback
- KU4.** the tools for roadmap creation and prioritization techniques (e.g., MoSCoW, Kano Model) and feature scoping
- KU5.** the use of prototyping and wireframing tools for visualizing early product features
- KU6.** how to analyze product metrics and trends
- KU7.** how to monitor product development activities
- KU8.** CI/CD pipeline tools and technologies (e.g., Jenkins, Travis CI)
- KU9.** build automation tools and scripting languages (e.g., Gradle, Maven)
- KU10.** best practices for automating deployment and testing processes to ensure rapid delivery

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** work independently and collaboratively
- GS3.** communicate effectively in writing
- GS4.** evaluate business impact and share relevant information
- GS5.** analyze data and assess its implications for the business
- GS6.** relay relevant information to others
- GS7.** maintain attention to detail
- GS8.** ensure work is complete and error-free
- GS9.** adhere to rule-based decision-making processes
- GS10.** plan and organize work to meet targets and deadlines
- GS11.** apply problem-solving approaches in various situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Build and test automation</i>	15	25	-	8
PC1. manage application environment variables and configurations for target environments	5	10	-	3
PC2. automate build and testing using scripts, tools, and test frameworks to ensure consistency and efficiency	5	10	-	3
PC3. integrate automated tests into the CI pipeline to detect errors early in the development cycle	5	5	-	2
<i>CI/CD pipeline management</i>	10	15	-	6
PC4. implement and manage CI/CD pipelines, ensuring a smooth and automated process for continuous integration and delivery	4	5	-	2
PC5. test, identify, and fix build failures to ensure uninterrupted development and deployment cycles	3	5	-	2
PC6. manage deployment policies and processes, ensuring security and compliance are adhered to in the deployment pipeline	3	5	-	2
<i>Deployment automation</i>	5	10	-	6
PC7. automate application deployment to different environments (e.g., web servers, API services, databases) using industry-standard tools (e.g., Jenkins, GitLab CI)	3	5	-	3
PC8. ensure deployment processes are scalable and maintain high availability, using containerization and orchestration tools (e.g. Docker, Kubernetes)	2	5	-	3
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8451
NOS Name	Implement Continuous Integration and Continuous Deployment (CI/CD) practices
Sector	IT-ITes
Sub-Sector	
Occupation	Web & Mobile Development
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N8125: Develop tests or simulations for end-to-end QA of systems

Description

This unit is about designing and developing test or simulation infrastructures to perform end-to-end QA of functionality, usability, compatibility, security, and/or performance of hardware and software systems that are supporting the deployment of algorithmic models.

Scope

The scope covers the following :

- Define requirements
- Create test cases
- Run test cases
- Conduct performance and load testing

Elements and Performance Criteria

Define requirements

To be competent, the user/individual on the job must be able to:

- PC1.** define functional requirements of the autonomous system, ensuring clarity and completeness
- PC2.** define the type of testing and testing requirements, including unit, sub-system, and system testing, based on project needs
- PC3.** identify issues with the requirements for testing and clarify these with relevant stakeholders
- PC4.** access reusable scenarios, test cases, scripts, and tools from the organization's knowledge base
- PC5.** integrate test cases into the CI/CD pipeline, ensuring smooth transitions from development to testing

Create test cases

To be competent, the user/individual on the job must be able to:

- PC6.** create or modify test scenarios relevant to the requirements, ensuring alignment with project goals
- PC7.** create or modify software test cases relevant to the requirements, focusing on functionality and performance
- PC8.** create or modify hardware test cases relevant to the requirements, covering all necessary specifications
- PC9.** identify test cases that can be automated feasibly, optimizing the testing process
- PC10.** create or modify automated scripts relevant to the requirements, improving efficiency in testing
- PC11.** access or create test data relevant to the requirements, ensuring comprehensive coverage
- PC12.** create a test plan to cover all requirements, detailing scope, objectives, and resources needed

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PC13. deploy code to lower environments for testing, leading up to the production environment, ensuring proper validation at each stage

Run test cases

To be competent, the user/individual on the job must be able to:

PC14. run the simulated test cases and evaluate the outcomes, providing accurate assessments of functionality

PC15. communicate the outcomes of the tests or simulations with relevant stakeholders and iterate based on feedback

PC16. create documentation on the tests or simulations for stakeholders, maintaining transparency and traceability

PC17. validate the test plan, test cases, and/or automated scripts with relevant stakeholders, ensuring all criteria are met

Conduct performance and load testing

To be competent, the user/individual on the job must be able to:

PC18. conduct performance and load testing, including advanced load, stress, and performance testing, aligning with industry practices

PC19. analyze test results to identify performance bottlenecks and areas for improvement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the purpose and aims of the autonomous system being developed, ensuring alignment with business objectives

KU2. organizational policies, procedures and priorities for developing tests or simulations for hardware and software systems

KU3. how to access and update information on scenarios, test cases, scripts, and tools in the organization's knowledge base

KU4. who to involve when developing tests or simulations for hardware and software systems, facilitating effective collaboration

KU5. organizational policies and procedures for documenting simulations, maintaining clear records

KU6. the application and use of test or simulation tools, understanding their capabilities and limitations

KU7. the approval process for testing software and hardware systems, ensuring compliance with standards

KU8. different sources of information available for designing tests and how to access these effectively

KU9. the differences between various types of testing, such as unit, sub-system, system, and automated testing

KU10. issues that may occur with testing requirements and how to address these proactively

KU11. the principles of designing tests, including best practices and methodologies

KU12. different simulation tool providers, such as Autodesk, and how to use their tools effectively

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- KU13.** how to evaluate and select the best simulation tool for specific purposes, considering project requirements
- KU14.** different test automation tools, such as Selenium, and how to use them in test scenarios
- KU15.** how to create suitable tests for particular applications, focusing on user needs and requirements
- KU16.** how to check whether test cases are suitable for automation, ensuring efficiency
- KU17.** how to access, create, and modify various high-level scenarios, test cases, automated scripts, test data, and test plans
- KU18.** different simulation methodologies and procedures used to ensure tests are effective and fit for purpose
- KU19.** how to analyze and use feedback to improve simulations, enhancing overall testing quality
- KU20.** scripting/programming languages to understand and implement test cases effectively
- KU21.** current practices in the design of tests for software and hardware systems, staying updated with industry trends
- KU22.** how to identify and refer anomalies during testing, ensuring timely resolution
- KU23.** how to work on various operating systems, such as Linux, Ubuntu, or Windows, adapting to different environments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make informed decisions on suitable courses of action
- GS2.** apply effective problem-solving approaches in diverse situations
- GS3.** analyze data and activities to derive meaningful insights
- GS4.** configure data and communicate relevant information effectively to others
- GS5.** apply balanced judgment to evaluate different situations
- GS6.** ensure work is thorough and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Define requirements</i>	8	15	-	5
PC1. define functional requirements of the autonomous system, ensuring clarity and completeness	2	4	-	1
PC2. define the type of testing and testing requirements, including unit, sub-system, and system testing, based on project needs	1	4	-	1
PC3. identify issues with the requirements for testing and clarify these with relevant stakeholders	2	3	-	1
PC4. access reusable scenarios, test cases, scripts, and tools from the organization's knowledge base	1	2	-	1
PC5. integrate test cases into the CI/CD pipeline, ensuring smooth transitions from development to testing	2	2	-	1
<i>Create test cases</i>	12	20	-	8
PC6. create or modify test scenarios relevant to the requirements, ensuring alignment with project goals	1	3	-	1
PC7. create or modify software test cases relevant to the requirements, focusing on functionality and performance	2	3	-	1
PC8. create or modify hardware test cases relevant to the requirements, covering all necessary specifications	1	3	-	1
PC9. identify test cases that can be automated feasibly, optimizing the testing process	2	2	-	1
PC10. create or modify automated scripts relevant to the requirements, improving efficiency in testing	1	3	-	1
PC11. access or create test data relevant to the requirements, ensuring comprehensive coverage	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. create a test plan to cover all requirements, detailing scope, objectives, and resources needed	2	2	-	1
PC13. deploy code to lower environments for testing, leading up to the production environment, ensuring proper validation at each stage	2	2	-	1
<i>Run test cases</i>	7	10	-	5
PC14. run the simulated test cases and evaluate the outcomes, providing accurate assessments of functionality	2	3	-	2
PC15. communicate the outcomes of the tests or simulations with relevant stakeholders and iterate based on feedback	1	2	-	1
PC16. create documentation on the tests or simulations for stakeholders, maintaining transparency and traceability	2	2	-	1
PC17. validate the test plan, test cases, and/or automated scripts with relevant stakeholders, ensuring all criteria are met	2	3	-	1
<i>Conduct performance and load testing</i>	3	5	-	2
PC18. conduct performance and load testing, including advanced load, stress, and performance testing, aligning with industry practices	1.5	3	-	1
PC19. analyze test results to identify performance bottlenecks and areas for improvement	1.5	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8125
NOS Name	Develop tests or simulations for end-to-end QA of systems
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N8418: Fix application bugs and improve application performance

Description

This unit is about identify bugs in an application, isolating them and then analyzing the logs to ensure smooth application performance

Scope

The scope covers the following :

- Record and document bugs
- Identify the bug
- Isolate the bug
- Analyze the logs

Elements and Performance Criteria

Record and document bugs

To be competent, the user/individual on the job must be able to:

- PC1.** Record the bug or enter it in the case tracking system
- PC2.** identify what the user was doing, what they were expecting, and what happened
- PC3.** search for relevant solutions for the error on developer forums
- PC4.** document the bug's impact, steps to reproduce it, and any attempted solutions
- PC5.** maintain detailed records of error messages, their respective solutions, and changes made during the debugging process

Identify the bug

To be competent, the user/individual on the job must be able to:

- PC6.** determine the immediate line of code where the bug occurs
- PC7.** specify the bug type (e.g., unexpected null, bad input, off-by-one, buffer overflow, index out-of-range, etc.)
- PC8.** document where the error occurs and note any relevant patterns in the code that might suggest a recurring issue

Isolate the bug

To be competent, the user/individual on the job must be able to:

- PC9.** use the process of elimination to isolate the bug to a particular line of code
- PC10.** disable blocks of code (comment them out) until the crash stops happening
- PC11.** use a unit-testing framework to isolate methods
- PC12.** continue to disable code and reduce the application to minimal functionality until it begins working again
- PC13.** eliminate the hardware or platform as a cause
- PC14.** log which parts of the code were disabled, tested, and how the behaviour changed

Analyze the logs

To be competent, the user/individual on the job must be able to:

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- PC15.** log all activities and analyze the logs
- PC16.** continue the isolation and logging processes until the immediate line of code where the bug occurs is identified
- PC17.** document log analysis results, noting which logs helped pinpoint the issue and which didn't

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the procedures and guidelines for analyzing functional and non-functional requirements for the solution
- KU2.** the procedures and guidelines related to logging and log analysis
- KU3.** the policies and procedures for sharing data
- KU4.** the policies and procedures for documenting results of application tests and log analysis
- KU5.** the relevant stakeholders to be involved for identifying bugs in the application code
- KU6.** the effective use of standard templates and tools
- KU7.** how to address the common issues that may occur during testing
- KU8.** how to analyze application log files
- KU9.** different test automation tools and how to use them
- KU10.** how to create suitable tests for particular applications
- KU11.** how to access, create, and modify different types of - high-level scenarios, test cases, automatic scripts, test data and test plans
- KU12.** the scripting/ programming of languages to understand test cases
- KU13.** how to identify and record bugs in the application code
- KU14.** how to analyze and use feedback to fix application bugs
- KU15.** how to document error reports and solutions applied, ensuring future reference for similar issues
- KU16.** the best practices for maintaining error records and communicating them to team members or stakeholders

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** adhere to given protocols and standards to ensure accurate task completion
- GS2.** ask relevant individuals for guidance when unclear, ensuring correct understanding
- GS3.** actively listen and provide clear, accurate information to ensure mutual understanding
- GS4.** apply established rules and procedures to make consistent, informed decisions
- GS5.** prioritize tasks effectively to meet deadlines while maintaining quality
- GS6.** select the best course of action based on available information and analysis
- GS7.** quickly escalate discrepancies to the appropriate authority for resolution
- GS8.** consult others for advice or solutions when encountering challenges
- GS9.** evaluate data and activities logically to identify trends and resolve issues

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- GS10.** communicate necessary details promptly to keep stakeholders informed
- GS11.** make fair, well-considered decisions by weighing all relevant factors
- GS12.** ensure accuracy by focusing on critical details to prevent errors
- GS13.** review work thoroughly to verify completeness and correctness
- GS14.** manage tasks independently while contributing effectively in team settings

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Record and document bugs</i>	8	14	-	5
PC1. Record the bug or enter it in the case tracking system	2	4	-	1
PC2. identify what the user was doing, what they were expecting, and what happened	1	3	-	1
PC3. search for relevant solutions for the error on developer forums	2	3	-	1
PC4. document the bug's impact, steps to reproduce it, and any attempted solutions	1	2	-	1
PC5. maintain detailed records of error messages, their respective solutions, and changes made during the debugging process	2	2	-	1
<i>Identify the bug</i>	6	10	-	5
PC6. determine the immediate line of code where the bug occurs	2	3	-	2
PC7. specify the bug type (e.g., unexpected null, bad input, off-by-one, buffer overflow, index out-of-range, etc.)	2	4	-	2
PC8. document where the error occurs and note any relevant patterns in the code that might suggest a recurring issue	2	3	-	1
<i>Isolate the bug</i>	9	16	-	6
PC9. use the process of elimination to isolate the bug to a particular line of code	1	3	-	1
PC10. disable blocks of code (comment them out) until the crash stops happening	1	2	-	1
PC11. use a unit-testing framework to isolate methods	2	3	-	1
PC12. continue to disable code and reduce the application to minimal functionality until it begins working again	2	3	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. eliminate the hardware or platform as a cause	1	3	-	1
PC14. log which parts of the code were disabled, tested, and how the behaviour changed	2	2	-	1
<i>Analyze the logs</i>	7	10	-	4
PC15. log all activities and analyze the logs	1	3	-	2
PC16. continue the isolation and logging processes until the immediate line of code where the bug occurs is identified	3	4	-	1
PC17. document log analysis results, noting which logs helped pinpoint the issue and which didn't	3	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8418
NOS Name	Fix application bugs and improve application performance
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	5
Credits	1
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N8323: Monitor and manage cloud applications and the deployed systems

Description

This unit is about monitoring and managing the performance of the applications and the deployed systems.

Scope

The scope covers the following :

- Gather requirements
- Monitor system performance
- Report application performance

Elements and Performance Criteria

Gather requirements

To be competent, the user/individual on the job must be able to:

- PC1.** identify the factors which impact the performance of the cloud application
- PC2.** analyze ways to optimize applications in terms of cost and resource utilization

Monitor system performance

To be competent, the user/individual on the job must be able to:

- PC3.** define metrics to monitor application performance and the health of deployed systems
- PC4.** monitor application log reports for errors and clues about problems with the application and the deployed systems on the cloud
- PC5.** assess and deploy appropriate application monitoring tools to track cloud application performance
- PC6.** perform analysis to generate consumable reports about cloud application performance

Report application performance

To be competent, the user/individual on the job must be able to:

- PC7.** share application performance reports with relevant stakeholders
- PC8.** provide actionable insights for re-engineering the application

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to the monitoring of applications and deployed systems
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting performance insights of applications and deployed systems



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- KU4.** who to involve in sharing reports on the performance of applications and deployed systems
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different ways to monitor and optimize utilization of resources
- KU7.** how to define metrics to monitor applications and deployed systems
- KU8.** different types of application monitoring tools
- KU9.** how to interpret application performance reports

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ask for clarification and advice from appropriate people
- GS2.** listen effectively and communicate information accurately
- GS3.** work independently and collaboratively
- GS4.** analyze the impact of architectural and design decisions on business and organization
- GS5.** pass on relevant information to others
- GS6.** apply good attention to detail
- GS7.** work effectively in a customer-facing environment
- GS8.** plan and organize your work to achieve targets and meet deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Gather requirements</i>	7	12	-	5
PC1. identify the factors which impact the performance of the cloud application	3.5	6	-	2.5
PC2. analyze ways to optimize applications in terms of cost and resource utilization	3.5	6	-	2.5
<i>Monitor system performance</i>	16	26	-	10
PC3. define metrics to monitor application performance and the health of deployed systems	4	6.5	-	2.5
PC4. monitor application log reports for errors and clues about problems with the application and the deployed systems on the cloud	4	6.5	-	2.5
PC5. assess and deploy appropriate application monitoring tools to track cloud application performance	4	6.5	-	2.5
PC6. perform analysis to generate consumable reports about cloud application performance	4	6.5	-	2.5
<i>Report application performance</i>	7	12	-	5
PC7. share application performance reports with relevant stakeholders	3.5	6	-	2.5
PC8. provide actionable insights for re-engineering the application	3.5	6	-	2.5
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8323
NOS Name	Monitor and manage cloud applications and the deployed systems
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	1
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8452: Optimize Revenue, Security, and Accessibility in Web and Mobile Products

Description

This unit is about defining the key competencies required for optimizing revenue generation, ensuring robust product security, protecting data privacy, and enhancing accessibility in web and mobile products.

Scope

The scope covers the following :

- Revenue Strategy
- Product Solution Strategy
- Customer Requirement Analysis
- Product Security
- Data Privacy and Security
- Accessibility in Web and Mobile Products

Elements and Performance Criteria

Revenue Strategy

To be competent, the user/individual on the job must be able to:

- PC1.** develop and implement monetization models (e.g., subscriptions, ads, freemium models) based on market analysis and revenue opportunities
- PC2.** conduct A/B testing and use analytics tools to evaluate product performance and optimize revenue potential

Product Solution Strategy

To be competent, the user/individual on the job must be able to:

- PC3.** design and prioritize product features aligned with business goals, ROI, and customer needs
- PC4.** coordinate with cross-functional teams to ensure timely and efficient delivery of product solutions

Customer Requirement Analysis

To be competent, the user/individual on the job must be able to:

- PC5.** conduct surveys, interviews, and usability testing to gather customer insights and establish continuous feedback loops for iterative improvements
- PC6.** translate customer requirements into actionable product specifications

Product Security

To be competent, the user/individual on the job must be able to:

- PC7.** identify and address potential security threats and vulnerabilities through regular audits, penetration testing, and security protocols (e.g., encryption, secure authentication, API security)

Data Privacy and Security

To be competent, the user/individual on the job must be able to:

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- PC8.** adhere to global data privacy regulations (e.g., GDPR, CCPA) by establishing clear policies for data collection, storage, and usage
- PC9.** ensure compliance with data anonymization and encryption standards

Accessibility in Web and Mobile Products

To be competent, the user/individual on the job must be able to:

- PC10.** implement and audit accessibility guidelines (e.g., WCAG, ADA) to ensure compliance across web and mobile platforms
- PC11.** develop inclusive solutions, such as screen reader compatibility, voice commands, and alternative input methods

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the various monetization models (e.g., subscription-based, freemium, ad-supported) and techniques for analyzing market opportunities and user behavior patterns to optimize revenue
- KU2.** the product lifecycle management and prioritization frameworks (e.g., MoSCoW, RICE) to guide feature development and iterative improvements
- KU3.** the methods for collecting and analyzing customer insights (e.g., surveys, interviews) and translating them into actionable product specifications using tools like Jira or Trello
- KU4.** the common security threats (e.g., phishing, SQL injection) and the application of secure coding practices, vulnerability testing tools, and compliance with standards such as ISO 27001
- KU5.** the global data privacy laws (e.g., GDPR, CCPA) and principles of data anonymization, encryption, and secure API integration for maintaining transparency and compliance
- KU6.** the principles of accessibility guidelines (e.g., WCAG, ADA) and inclusive design practices, supported by tools like Lighthouse, to ensure usability for diverse user groups
- KU7.** the digital marketing strategies and emerging trends in web and mobile technologies (e.g., AI, AR/VR) to align product solutions with market demands
- KU8.** the integration of analytics tools and dashboards to monitor KPIs related to revenue, security, and accessibility for data-driven decision-making
- KU9.** the project management methodologies (e.g., Agile, Scrum) and effective communication techniques to coordinate cross-functional teams and stakeholders
- KU10.** the customer-centric approaches and problem-solving techniques to address complex product challenges and prioritize user needs in development strategies

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and advice from relevant individuals when needed
- GS2.** listen actively and communicate information clearly and accurately
- GS3.** adhere to structured decision-making frameworks and guidelines
- GS4.** make well-informed decisions based on available data and context



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- GS5.** plan tasks effectively to meet targets and deadlines
- GS6.** apply problem-solving techniques to address various situations
- GS7.** organize and share relevant information clearly with others
- GS8.** use balanced judgment to evaluate situations and make decisions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Revenue Strategy</i>	5	10	-	4
PC1. develop and implement monetization models (e.g., subscriptions, ads, freemium models) based on market analysis and revenue opportunities	3	5	-	2
PC2. conduct A/B testing and use analytics tools to evaluate product performance and optimize revenue potential	2	5	-	2
<i>Product Solution Strategy</i>	5	10	-	4
PC3. design and prioritize product features aligned with business goals, ROI, and customer needs	3	5	-	2
PC4. coordinate with cross-functional teams to ensure timely and efficient delivery of product solutions	2	5	-	2
<i>Customer Requirement Analysis</i>	5	10	-	3
PC5. conduct surveys, interviews, and usability testing to gather customer insights and establish continuous feedback loops for iterative improvements	3	5	-	2
PC6. translate customer requirements into actionable product specifications	2	5	-	1
<i>Product Security</i>	5	5	-	3
PC7. identify and address potential security threats and vulnerabilities through regular audits, penetration testing, and security protocols (e.g., encryption, secure authentication, API security)	5	5	-	3
<i>Data Privacy and Security</i>	5	10	-	3
PC8. adhere to global data privacy regulations (e.g., GDPR, CCPA) by establishing clear policies for data collection, storage, and usage	3	5	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. ensure compliance with data anonymization and encryption standards	2	5	-	1
<i>Accessibility in Web and Mobile Products</i>	5	5	-	3
PC10. implement and audit accessibility guidelines (e.g., WCAG, ADA) to ensure compliance across web and mobile platforms	3	3	-	2
PC11. develop inclusive solutions, such as screen reader compatibility, voice commands, and alternative input methods	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8452
NOS Name	Optimize Revenue, Security, and Accessibility in Web and Mobile Products
Sector	IT-ITes
Sub-Sector	
Occupation	Web & Mobile Development
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SSC/N8414: Develop a user-friendly web app aligned with functional, non-functional, and UX requirements

Description

This unit is about developing web applications for different platforms keeping in mind the different requirements of the application/ solution.

Scope

The scope covers the following :

- Understanding the scope
- Design and development

Elements and Performance Criteria

Understanding the scope

To be competent, the user/individual on the job must be able to:

- PC1.** Collaborate with cross functional teams to understand the scope
- PC2.** Understand and analysed the functional, non-functional and user experience requirements with which the interface must be developed
- PC3.** Create list of tasks that the user can execute within the interface based on the requirements identified
- PC4.** Organise the list of tasks and interfaces needed for the overall application
- PC5.** Create a pre-list of possible re-usable components before starting the development
- PC6.** use Git/GitHub for version control to manage code changes and collaboration in a team environment

Design and development

To be competent, the user/individual on the job must be able to:

- PC7.** define the structure of the pages, including headers, sections, articles, main, and footer
- PC8.** develop codes for the various components such as headers, sections, articles, main, and footer
- PC9.** write application code that meets security requirements
- PC10.** design and develop unit tests for the application code
- PC11.** build, test, and run the application before deployment
- PC12.** manage data storage and retrieval mechanisms, ensuring efficiency and security
- PC13.** conduct accessibility testing to ensure the web application is usable by all individuals, including those with disabilities
- PC14.** apply agile methodologies like Scrum or Kanban to manage the development cycle and ensure continuous feedback and iteration

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** the procedures and guidelines for building and maintaining applications
- KU2.** the procedures and guidelines for sharing data and managing security configurations
- KU3.** the stakeholders to be involved when defining continuous integration and continuous delivery practices
- KU4.** the effective use of standard templates and tools
- KU5.** how to deploy application dependencies and manage data storage
- KU6.** how to create serverless applications and microservices
- KU7.** how to develop service-oriented architecture-based applications and secure APIs
- KU8.** how to scale applications horizontally and build resilient applications
- KU9.** different container technologies and how to build applications from source code
- KU10.** how to test applications for build failures and automate the build and testing process
- KU11.** different tools for automating application build and testing
- KU12.** the use of Git/GitHub for version control in a collaborative team environment
- KU13.** the principles of data storage and management, including backup and retrieval processes
- KU14.** different security standards, authentication measures, and compliance/regulatory standards
- KU15.** how to conduct accessibility testing to ensure web applications meet accessibility standards
- KU16.** the basics of agile methodologies, including Scrum and Kanban, and their role in the development process
- KU17.** how to encrypt data to maintain confidentiality and security

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow established protocols to ensure tasks are completed accurately and in compliance with standards
- GS2.** manage tasks autonomously while effectively collaborating with team members when needed
- GS3.** convey information clearly and concisely in written form, ensuring proper understanding by the recipient
- GS4.** comprehend how architectural decisions affect the business and organizational structure to align development with broader goals
- GS5.** analyze how decisions or changes impact the business and ensure that key information is communicated to relevant stakeholders
- GS6.** provide timely and necessary information to colleagues or stakeholders to support decision-making and task completion
- GS7.** review and verify that tasks are finished thoroughly and meet quality standards without mistakes
- GS8.** use a balanced and thoughtful approach to make fair decisions, considering all relevant factors
- GS9.** select appropriate actions based on available data, options, and situational analysis



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- GS10.** efficiently schedule tasks and manage resources to ensure objectives are achieved within specified timeframes
- GS11.** employ logical and creative thinking to identify solutions and resolve issues in different contexts
- GS12.** proactively seek advice or clarification from others when confronted with challenges or uncertainties

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the scope</i>	13	24	-	8
PC1. Collaborate with cross functional teams to understand the scope	2	4	-	1.5
PC2. Understand and analysed the functional, non-functional and user experience requirements with which the interface must be developed	3	4	-	1.5
PC3. Create list of tasks that the user can execute within the interface based on the requirements identified	2	4	-	1.5
PC4. Organise the list of tasks and interfaces needed for the overall application	2	4	-	1.5
PC5. Create a pre-list of possible re-usable components before starting the development	2	4	-	1
PC6. use Git/GitHub for version control to manage code changes and collaboration in a team environment	2	4	-	1
<i>Design and development</i>	17	26	-	12
PC7. define the structure of the pages, including headers, sections, articles, main, and footer	2	3	-	1.5
PC8. develop codes for the various components such as headers, sections, articles, main, and footer	2	4	-	1.5
PC9. write application code that meets security requirements	2	3	-	1.5
PC10. design and develop unit tests for the application code	3	4	-	1.5
PC11. build, test, and run the application before deployment	2	3	-	1.5
PC12. manage data storage and retrieval mechanisms, ensuring efficiency and security	2	3	-	1.5



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. conduct accessibility testing to ensure the web application is usable by all individuals, including those with disabilities	2	3	-	1.5
PC14. apply agile methodologies like Scrum or Kanban to manage the development cycle and ensure continuous feedback and iteration	2	3	-	1.5
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8414
NOS Name	Develop a user-friendly web app aligned with functional, non-functional, and UX requirements
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8415: Develop native/cross-platform/hybrid mobile application for the target platforms

Description

This unit is about developing mobile applications for different types of mobile platforms

Scope

The scope covers the following :

- Requirements mapping
- Application development
- Publishing to app-store
- Independent application development

Elements and Performance Criteria

Requirements mapping

To be competent, the user/individual on the job must be able to:

- PC1.** obtain information about the mobile solution, user needs, and market trends
- PC2.** identify the functional, non-functional, and user experience (UX) requirements of the mobile application
- PC3.** identify dependencies related to development, including device hardware functionalities, third-party integrations, and time-to-market
- PC4.** follow emerging technologies in mobile app development, such as AI integration, augmented reality (AR), and machine learning (ML) to incorporate them into planning

Application development

To be competent, the user/individual on the job must be able to:

- PC5.** develop application code according to security and compliance requirements
- PC6.** encrypt data to safeguard user information where applicable
- PC7.** manage security configurations to meet regulatory standards
- PC8.** design and develop unit tests to ensure code reliability
- PC9.** build, run, and test the application before deployment
- PC10.** develop platform capabilities for A/B testing of features
- PC11.** integrate AI, AR, or ML technologies to enhance app functionality and user engagement
- PC12.** design user-friendly interfaces that enhance the overall user experience (UX/UI)

Publishing to app-stores

To be competent, the user/individual on the job must be able to:

- PC13.** create and maintain service configurations for code deployment
- PC14.** automate the deployment process using scripts and tools for efficiency
- PC15.** publish the mobile application to the relevant app stores or platforms, ensuring adherence to their guidelines

Qualification Pack

PC16. manage updates and continuous delivery using best practices in app-store deployment

Independent application development

To be competent, the user/individual on the job must be able to:

PC17. develop and manage mobile applications independently, from concept to deployment

PC18. handle all aspects of the app development lifecycle, including requirement gathering, coding, testing, and app store publishing

PC19. follow market trends and new technologies to continuously improve application features and performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the procedures for building, maintaining, and securing applications

KU2. how to develop native, hybrid, and cross-platform mobile applications

KU3. the impact of emerging technologies like AI, AR, and ML in mobile app development

KU4. the principles of designing user-friendly and visually appealing UX/UI interfaces

KU5. how to encrypt and obfuscate data on mobile devices to maintain data security

KU6. different types of security standards, protocols, and authentication measures

KU7. the processes for automating the build, testing, and deployment of mobile applications

KU8. how to navigate app-store regulations, publish apps, and maintain updates

KU9. tools and techniques for scaling mobile applications horizontally and ensuring resilience

KU10. best practices for independent app development, including managing multiple roles

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. adhere to given protocols and standards to ensure accurate task completion

GS2. ask relevant individuals for guidance when unclear, ensuring correct understanding

GS3. actively listen and provide clear, accurate information to ensure mutual understanding

GS4. apply established rules and procedures to make consistent, informed decisions

GS5. prioritize tasks effectively to meet deadlines while maintaining quality

GS6. select the best course of action based on available information and analysis

GS7. quickly escalate discrepancies to the appropriate authority for resolution

GS8. consult others for advice or solutions when encountering challenges

GS9. evaluate data and activities logically to identify trends and resolve issues

GS10. communicate necessary details promptly to keep stakeholders informed

GS11. make fair, well-considered decisions by weighing all relevant factors

GS12. ensure accuracy by focusing on critical details to prevent errors

GS13. review work thoroughly to verify completeness and correctness

GS14. manage tasks independently while contributing effectively in team settings

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Requirements mapping</i>	6	13	-	4
PC1. obtain information about the mobile solution, user needs, and market trends	2	4	-	1
PC2. identify the functional, non-functional, and user experience (UX) requirements of the mobile application	1	4	-	1
PC3. identify dependencies related to development, including device hardware functionalities, third-party integrations, and time-to-market	2	3	-	1
PC4. follow emerging technologies in mobile app development, such as AI integration, augmented reality (AR), and machine learning (ML) to incorporate them into planning	1	2	-	1
<i>Application development</i>	12	20	-	8
PC5. develop application code according to security and compliance requirements	2	2	-	1
PC6. encrypt data to safeguard user information where applicable	1	3	-	1
PC7. manage security configurations to meet regulatory standards	2	3	-	1
PC8. design and develop unit tests to ensure code reliability	1	3	-	1
PC9. build, run, and test the application before deployment	2	2	-	1
PC10. develop platform capabilities for A/B testing of features	1	3	-	1
PC11. integrate AI, AR, or ML technologies to enhance app functionality and user engagement	1	2	-	1
PC12. design user-friendly interfaces that enhance the overall user experience (UX/UI)	2	2	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Publishing to app-stores</i>	7	9	-	5
PC13. create and maintain service configurations for code deployment	2	2	-	1
PC14. automate the deployment process using scripts and tools for efficiency	2	3	-	2
PC15. publish the mobile application to the relevant app stores or platforms, ensuring adherence to their guidelines	1	2	-	1
PC16. manage updates and continuous delivery using best practices in app-store deployment	2	2	-	1
<i>Independent application development</i>	5	8	-	3
PC17. develop and manage mobile applications independently, from concept to deployment	2	3	-	1
PC18. handle all aspects of the app development lifecycle, including requirement gathering, coding, testing, and app store publishing	1.5	3	-	1
PC19. follow market trends and new technologies to continuously improve application features and performance	1.5	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8415
NOS Name	Develop native/cross-platform/hybrid mobile application for the target platforms
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8416: Develop reliable, scalable and secure back-end aligned to the application architecture

Description

This unit is about developing reliable and secure back-end APIs and services for any software product/solution.

Scope

The scope covers the following :

- Identification of scope
- Design back-end
- Design back-end APIs and services
- Design database
- Reliability and scalability
- Write test scripts
- Back-end development

Elements and Performance Criteria

Identification of scope

To be competent, the user/individual on the job must be able to:

- PC1.** demarcate backend and frontend responsibilities before starting application development
- PC2.** identify the scope of backend operations and functionalities, ensuring alignment with project requirements
- PC3.** consider the implications of different Software Development Life Cycle (SDLC) models, such as Agile, Waterfall, and DevOps, on backend development planning and execution

Design back-end

To be competent, the user/individual on the job must be able to:

- PC4.** design server endpoints to connect with the client efficiently
- PC5.** create stubs for upstream and downstream components to facilitate backend functionality

Design back-end APIs and services

To be competent, the user/individual on the job must be able to:

- PC6.** document what each backend API endpoint must accomplish and its purpose
- PC7.** specify the required and optional values expected from the client and those returned by the backend
- PC8.** ensure the documentation is accurate and consistently updated
- PC9.** implement retry logic and circuit breakers to handle failures in backend systems and ensure smooth operation

Design database

To be competent, the user/individual on the job must be able to:

- PC10.** design and document the database schema, considering scalability and performance needs

Qualification Pack

Reliability and scalability

To be competent, the user/individual on the job must be able to:

- PC11.** ensure all processes align with reliability and scalability requirements
- PC12.** apply micro services architecture to enhance scalability, fault isolation, and ease of maintenance

Write test scripts

To be competent, the user/individual on the job must be able to:

- PC13.** write test scripts to verify the functionality and reliability of all backend endpoints

Back-end development

To be competent, the user/individual on the job must be able to:

- PC14.** Build the API using a selected programming language
- PC15.** Deploy backend using a cloud service or a dedicated host

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the procedures and guidelines for building and maintaining applications
- KU2.** the best practices for sharing data and managing security configurations within an organization
- KU3.** the stakeholders to be involved while defining continuous integration and delivery (CI/CD) practices
- KU4.** various templates, tools, and resources available for backend development
- KU5.** different models of SDLC, such as Agile, Waterfall, and DevOps, and their impact on backend development
- KU6.** the benefits and structure of microservices architecture for scalability and maintainability
- KU7.** the various types of backend software architectures and frameworks
- KU8.** how to develop secure backend APIs and services that comply with industry standards
- KU9.** strategies for handling failures in backend systems, including retry logic, circuit breakers, and failover mechanisms
- KU10.** how to horizontally scale applications to meet growing user demands
- KU11.** how to automate backend tasks using scripts and tools
- KU12.** automation of the application build and testing process using available tools
- KU13.** different security standards, protocols, and authentication measures
- KU14.** how to encrypt and obfuscate data on mobile devices for security purposes
- KU15.** how to monitor server performance and resource usage for optimal operation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** Work independently and collaboratively



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- GS3.** Communicate with others in writing
- GS4.** evaluate the business impact of architecture decisions
- GS5.** share relevant business impact information with others
- GS6.** communicate the necessary information clearly
- GS7.** ensure work is accurate and complete
- GS8.** apply sound judgment in different scenarios
- GS9.** make informed decisions on suitable actions
- GS10.** plan and organize tasks to meet deadlines
- GS11.** use problem-solving approaches as needed
- GS12.** seek clarification on issues when necessary

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identification of scope</i>	6	11	-	4
PC1. demarcate backend and frontend responsibilities before starting application development	2	4	-	1
PC2. identify the scope of backend operations and functionalities, ensuring alignment with project requirements	2	4	-	1.5
PC3. consider the implications of different Software Development Life Cycle (SDLC) models, such as Agile, Waterfall, and DevOps, on backend development planning and execution	2	3	-	1.5
<i>Design back-end</i>	4	8	-	2
PC4. design server endpoints to connect with the client efficiently	2	4	-	1
PC5. create stubs for upstream and downstream components to facilitate backend functionality	2	4	-	1
<i>Design back-end APIs and services</i>	8	15	-	5
PC6. document what each backend API endpoint must accomplish and its purpose	2	4	-	1
PC7. specify the required and optional values expected from the client and those returned by the backend	2	4	-	1.5
PC8. ensure the documentation is accurate and consistently updated	2	3	-	1.5
PC9. implement retry logic and circuit breakers to handle failures in backend systems and ensure smooth operation	2	4	-	1
<i>Design database</i>	2	3	-	1.5



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. design and document the database schema, considering scalability and performance needs	2	3	-	1.5
<i>Reliability and scalability</i>	4	5	-	3
PC11. ensure all processes align with reliability and scalability requirements	2	2.5	-	1.5
PC12. apply micro services architecture to enhance scalability, fault isolation, and ease of maintenance	2	2.5	-	1.5
<i>Write test scripts</i>	2	3	-	1.5
PC13. write test scripts to verify the functionality and reliability of all backend endpoints	2	3	-	1.5
<i>Back-end development</i>	4	5	-	3
PC14. Build the API using a selected programming language	2	2.5	-	1.5
PC15. Deploy backend using a cloud service or a dedicated host	2	2.5	-	1.5
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8416
NOS Name	Develop reliable, scalable and secure back-end aligned to the application architecture
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8417.Implement DevSecOps practices for secure application development	30	50	0	20	100	20
SSC/N8451.Implement Continuous Integration and Continuous Deployment (CI/CD) practices	30	50	0	20	100	15
SSC/N8125.Develop tests or simulations for end-to-end QA of systems	30	50	0	20	100	15
SSC/N8418.Fix application bugs and improve application performance	30	50	0	20	100	15
SSC/N8323.Monitor and manage cloud applications and the deployed systems	30	50	0	20	100	15
SSC/N8452.Optimize Revenue, Security, and Accessibility in Web and Mobile Products	30	50	0	20	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	200	330	-	120	650	90

Elective: 1 Front-end Web Development

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8414.Develop a user-friendly web app aligned with functional, non-functional, and UX requirements	30	50	0	20	100	10
Total	30	50	-	20	100	10

Elective: 2 Mobile Application Development



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8415.Develop native/cross-platform/hybrid mobile application for the target platforms	30	50	0	20	100	10
Total	30	50	-	20	100	10

Elective: 3 Back-end Engineering

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8416.Develop reliable, scalable and secure back-end aligned to the application architecture	30	50	0	20	100	10
Total	30	50	-	20	100	10

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
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NSQF	National Skills Qualifications Framework
QP	Qualifications Pack

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
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Qualification Pack

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Qualification Pack

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Qualification Pack

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Organisational context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core skills/ Generic skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialised job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.